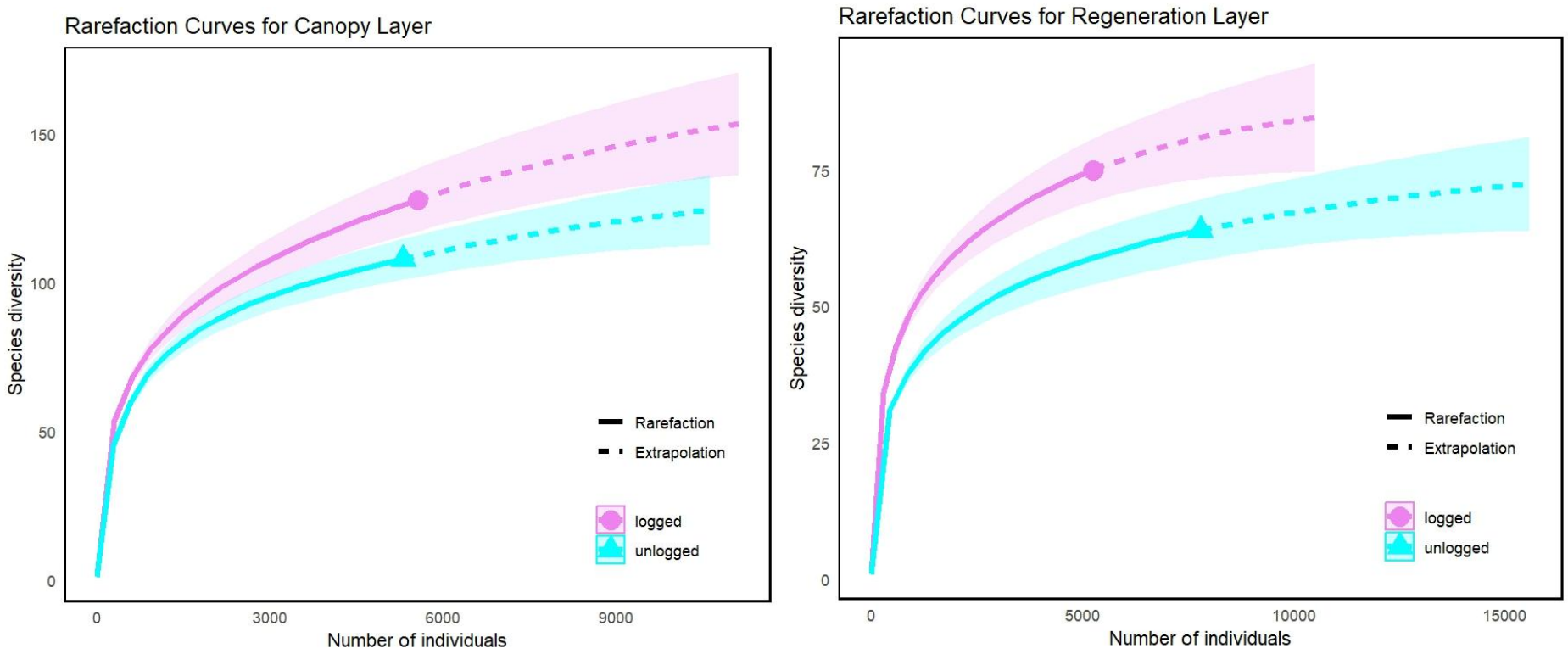
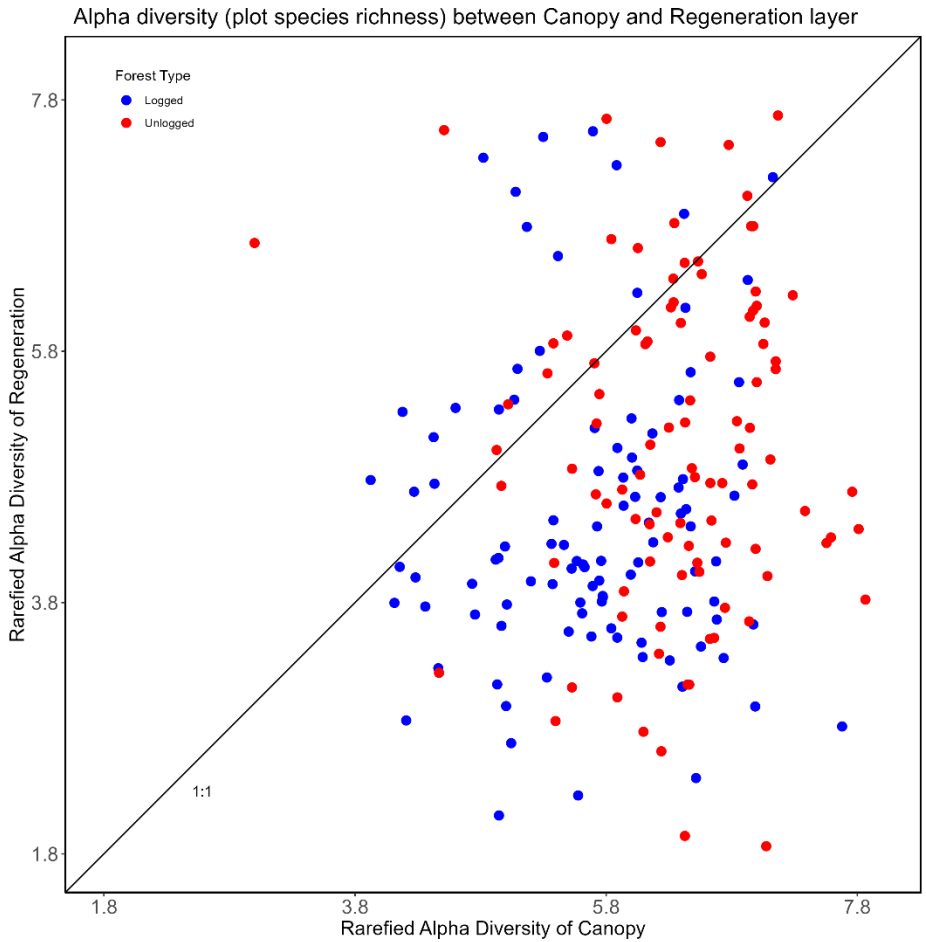
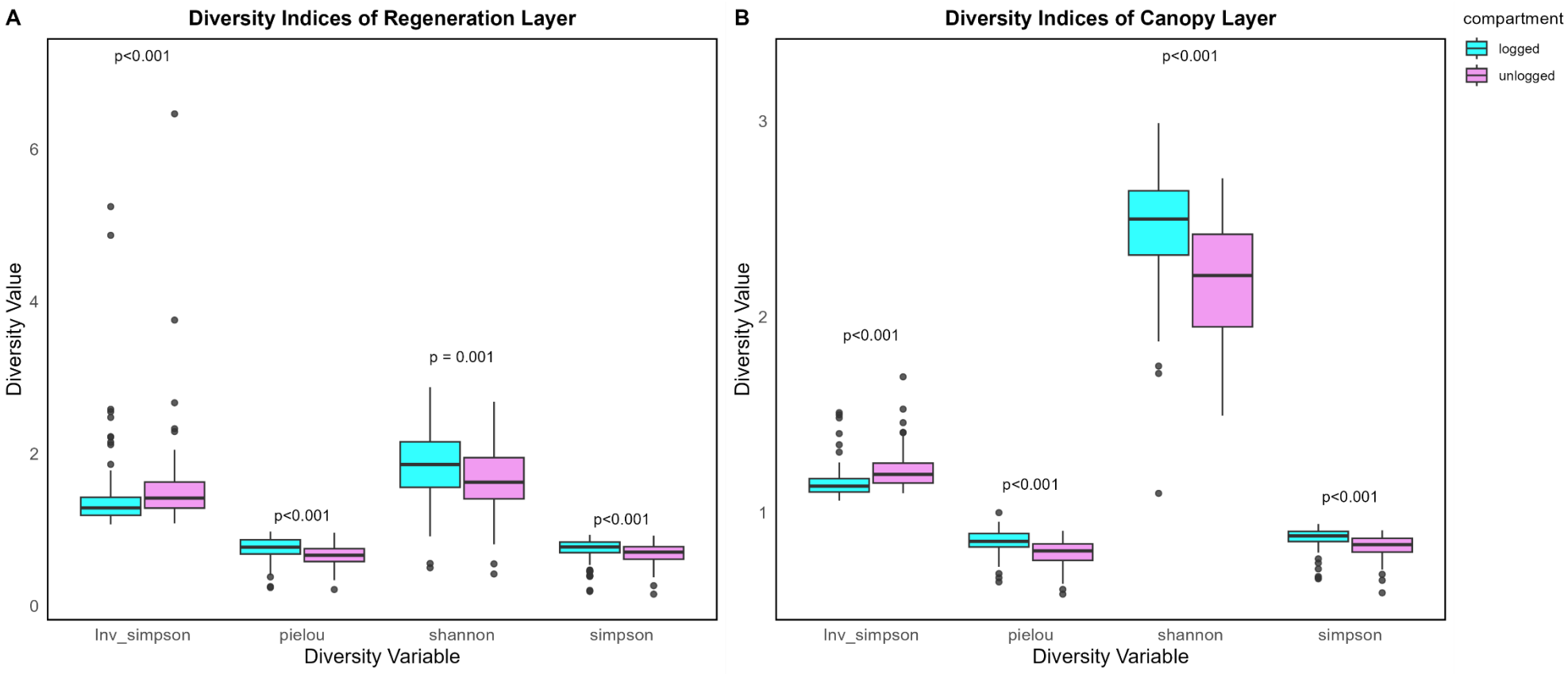
RESULTS



Rarefaction curves of species richness and abundance with 95% confidence intervals. Solid lines represent observed and interpolated species diversity, while dotted lines represent predicted species diversity based on extrapolation. Canopy and Regeneration curves are based on sample plots of $1000m^2$ and $16m^2$ respectively.



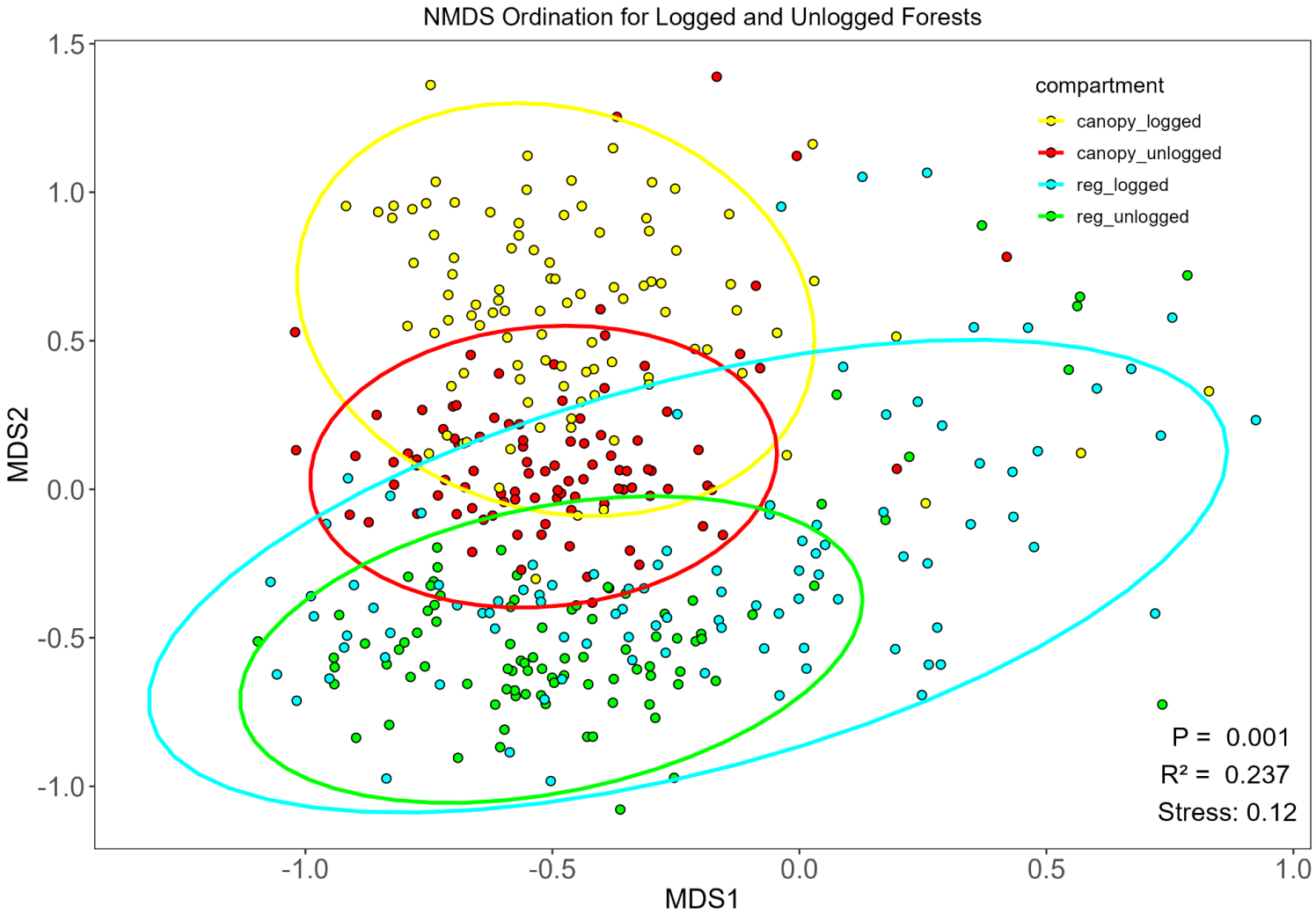
Rarefied alpha diversity between Canopy and Regeneration layer. The rarefied alpha diversity values are a measure of standardized species richness values for each plot (alpha diversity) and the diagonal 1:1 line depicts a slope of 1. A total of 204 sample plots for logged and unlogged forests were adopted.



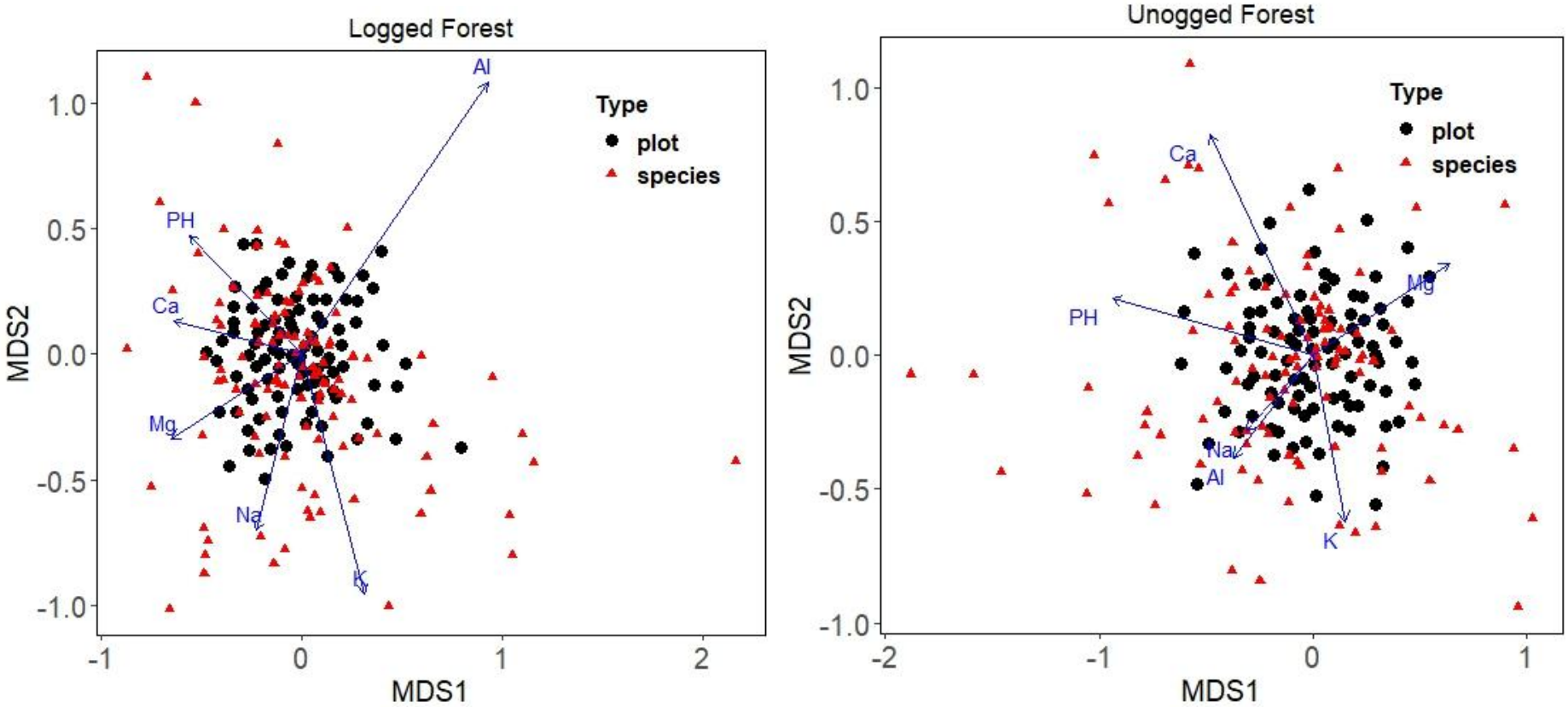
Diversity indices for regeneration and canopy layers between logged (cyan colour) and unlogged (purple colour) forest. Comparison of Pielou evenness, Shannon and Inverse Simpson diversity indices. P-value generated from ANOVA for plot means at a confidence interval of 95%. The plot presents the variability in the dataset for regeneration and Canopy layers based on 205 and 204 sample plots of $16m^2$ and $1000m^2$, respectively.

Statistics of the diversity indices of regeneration and canopy layers for logged and unlogged compartments within the Budongo Forest Reserve, Uganda. The mean and standard error (SE) are presented with F and p-values generated from Analysis of Variance (ANOVA) results at a confidence interval of 95% (0.05).

Category	*Variable*	*Logged forest mean (SE)*	*Unlogged forest mean (SE)*	*F value*	*P value*
Regeneration layer (DBH < 10cm)	Shannon index	1.84 ± 0.045	1.66 ± 0.043	8.931	0.0032
	Exponential Shannon	6.95 ± 0.29	5.74 ± 0.257	9.627	0.0022
	Simpson index	0.74 ± 0.014	0.69 ± 0.013	6.281	0.013
	Inverse Simpson	1.45 ± 0.058	1.54 ± 0.061	1.111	0.293
	Pielou index	0.75 ± 0.015	0.67 ± 0.014	15.07	0.00014
	Alpha diversity(α)	11.71 ± 0.268	12.14 ± 0.311	1.129	0.289
	Trees (N/m^2)	3.13 ± 0.17	4.869 ± 0.285	28.14	<0.000
	Gamma diversity(γ)	72	65	-	-
Canopy Layer (DBH ≥ 10cm)	Shannon index	2.47 ± 0.028	2.18 ± 0.03	49.35	<0.000
	Exponential Shannon	12.23 ± 0.302	9.24 ± 0.262	55.53	<0.000
	Simpson index	0.87 ± 0.005	0.83 ± 0.006	33.67	<0.000
	Inverse Simpson	1.15 ± 0.008	1.22 ± 0.01	26.97	<0.000
	Pielou index	0.85 ± 0.006	0.79 ± 0.007	44.21	<0.000
	Alpha diversity(α)	18.51 ± 0.359	15.85 ± 0.343	28.7	<0.000
	Trees (N/ha)	529.5 ± 13.60	536.6 ± 9.44	0.177	0.674
	Basal area(m^2/ha)	37.27 ± 1.319	39.87 ± 1.846	1.335	0.249
	AGB(kg/ha)	455.63 ± 23.48	576.95 ± 36.83	7.899	0.005
	Gamma diversity(γ)	128	108	-	-



Non-metric multidimensional Scaling (NMDS) ordination for the forest types and layers based on the sample plots. The ellipses are developed at a 95% confidence level for a multivariate t-distribution with overlapping areas illustrating similarity. Regeneration is represented as reg in the legend.



Non-metric multidimensional Scaling (NMDS) ordinations for Logged (left) and Unlogged (right) forests. Species abundances from both the canopy and regeneration layers were combined for each forest type to compute ordinations for species and plots. Mean concentrations of soil variables for each plot were fitted to the ordination to determine the influence of soil nutrients on the spatial distribution of species composition (plot) and individual species. The orientation of the arrow signifies the direction of the influence, while the length indicates the magnitude of the gradient or change. Inventory was carried out in systematically established permanent plots of 204 circular plots of 1,000 m^2 at 100m intervals along a 2.5 km transect, with each nested with four 4 m^2 subplots in selectively logged and unlogged forests.